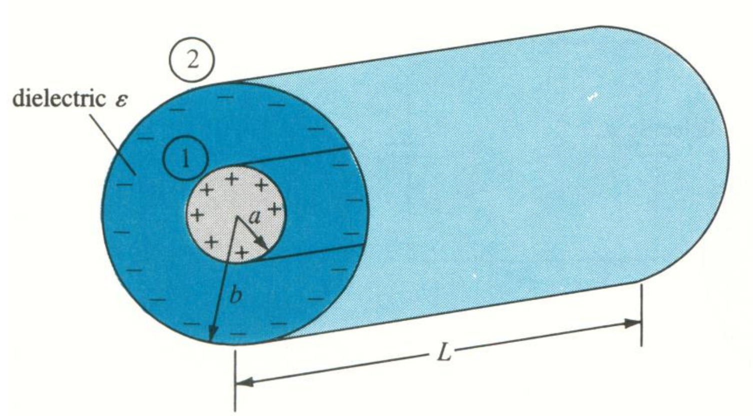
Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 6

1. Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\varepsilon_{r1} = 4$ and for $z \leq 0$, $\varepsilon_{r2} = 3$. A uniform electric field $\boldsymbol{E}_1 = 5\boldsymbol{a}_x - 2\boldsymbol{a}_y + 3\boldsymbol{a}_z\ kV/m$ exists for $z \geq 0$. Find:
   a. $\boldsymbol{E}_2$ for $z \leq 0$.
   b. The angles E_1 and E_2 make with the interface.
   c. The energy densities in J/m^3 in both dielectrics.
   d. The energy within a cube of side 2 m centered at $(3, 4, -5)$.

2. Consider the length L of two coaxial conductors of inner radius a and outer radius b (b>a) as shown in the following figure. Let the space between the conductors be filled with a homogeneous dielectric with permittivity ε. Find the capacitance of a coaxial cylinder.



3. Conducting spherical shells with radii a= 10 cm and b= 30 cm are maintained at a potential difference of 100 V such that V (r = b) = 0 and V (r = a) = 100 V. Determine V and **E** in the region between the shells. If $\varepsilon_r = 2.5$ in the region, determine the total charge induced on the shells and the capacitance of the system?